

Adham Ashraf Saeed

Assignment 4 (Final Project)

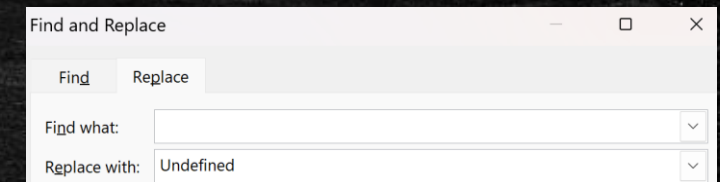
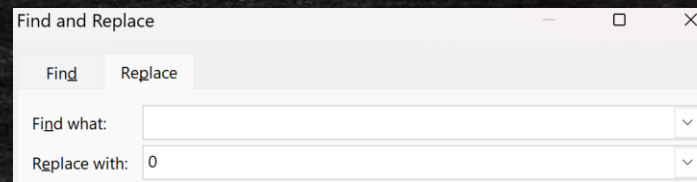
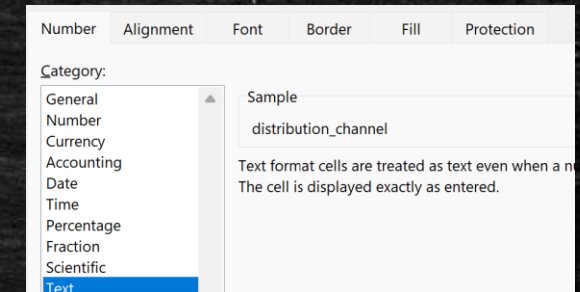
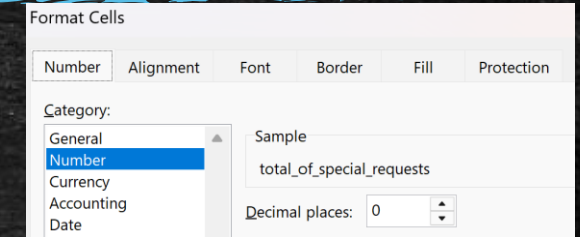
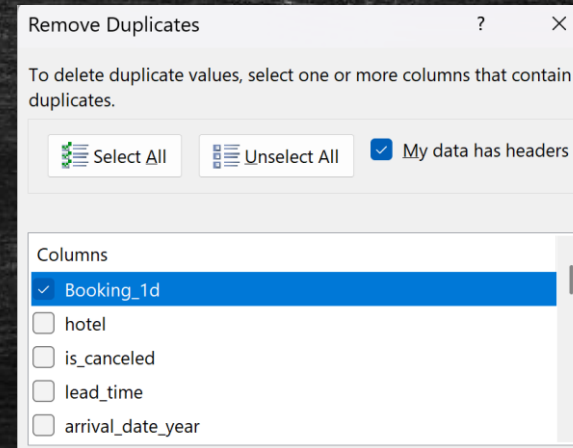
Project Objectives

- Clean and prepare raw data
- Use Excel formulas effectively
- Perform exploratory data analysis
- Build Pivot Tables & dashboards
- Generate business insights

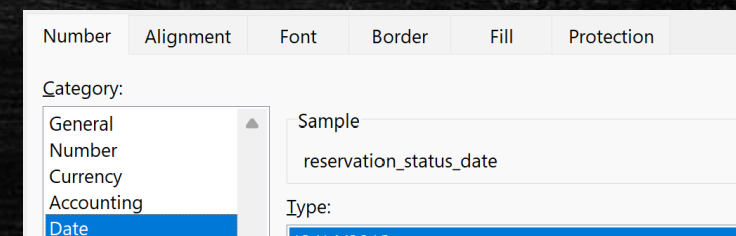
Task 1

Phase 1: Data Cleaning & Preparation

- Start with data Importing and Data Exploration and understanding each column and its values mean
- Remove duplicates (if any) In Primary Key(Booking_ID)
- Handle missing values
- Standardize text fields
- Identify Arrival Date Issue



Arrival_Date



```
=DATE(@arrival_date_year,MONTH(DATEVALUE(1&@arrival_date_month)),@arrival_date_day_of_month)
```


Task 2

Phase 2: Creating useful (Calculated Columns)

- Total Stay Nights
- Guest Type ("Single", "Couple", "Family")
- Arrival date (Month/ Day/ Year)
- Payment Risk("High Risk", "Low Risk") Based on Deposit Status
- Total Guests

Total Stay Nights

```
=@stays_in_weekend_nights+@stays_in_week_nights
```

Guest Type

```
=IF(@children+@babies>0,"Family",IF(@adults=1,"Single","Couple"))
```

Arrival_Date

```
=DATE(@arrival_date_year,MONTH(DATEVALUE(1&@arrival_date_month)),@arrival_date_day_of_month)
```

Payment Risk

```
=IF(@deposit_type="No Deposit","High Risk","Low Risk")
```

Total Guests

```
=@adults+@children+@babies
```


Task 2

Phase 2: Creating useful (Calculated Columns)

- Revenue per Booking
- Booking Status (Cancelled/ Not Cancelled)
- Lead Time Category (Long: Higher than 90 days , Medium, Short: less than 30)
- Feel Free to create more calculated columns to enhance your Analysis

Season

=SWITCH(@arrival_date_month,
"December","Winter",
"January","Winter",
"February","Winter",
"March","Spring",
"April","Spring",
"May","Spring",
"June","Summer",
"July","Summer",
"August","Summer",
"September","Autumn",
"October","Autumn",
"November","Autumn")

Revenue per Booking

=@Total_Stay_Nights*@Average_Daily_Rate

Booking Status

=IF(@is_canceled=1,"Cancelled","Not Cancelled")

Lead Time Category

=IF(@lead_time>90,"Long",IF(@lead_time<30,"Short","Medium"))

Revenue Category

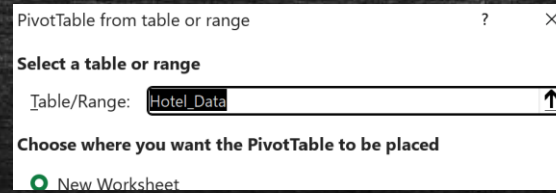
=IF(@Revenue_per_Booking>1500,"High",IF(@Revenue_per_Booking<500,"Low","Medium"))

Stay Type

=IF(@Total_Stay_Nights>5,"Long Stay",IF(@Total_Stay_Nights<3,"Short Stay","Medium Stay"))

Task 3

Phase 3: PivotTables & Insights



- Ask Strong business Questions and answer it by Pivot Tables

Question 1					Question 2						Question 3				
Which hotel type generates the highest revenue?					Which month has the highest number of bookings?					Which customer type has the highest cancellation rate?					
City Hotel					August						Transient				

Question 4						Question 5						Question 6									
Which country generates the highest number of bookings?						Which market segment generates the highest revenue?						Does lead time affect cancellations?									
PRT						Online TA						Yes. As lead time increases, the likelihood of cancellation also increases.									

Question 7				Question 8				Question 9			
Which guest type books the most?				Which season has the highest revenue?				What is the hotel booking cancellation rate?			
Couple				Summer				37.04%			

Task 4

Phase 4: Data Visualization

- Create charts to Visualize each
- Dashboard Creation: Build an interactive dashboard including:

KPIs:

1. Total Bookings
2. Total Revenue
3. Cancellation Rate

Total Revenue	Total Booking	Cancellation Rate
\$42,723,498	119,390	37.04%

Task 4

Phase 4: Data Visualization

- Create charts to Visualize each
- Dashboard Creation: Build an interactive dashboard including:

Filters (Slicers):

1. Hotel Type
2. Year
3. Customer Type

- Charts & PivotTables

hotel	arrival_date_year	customer_type
City Hotel	2015	Contract
Resort Hotel	2016	Group
	2017	Transient
		Transient-Party

Dashboard

Total Revenue

\$42,723,498

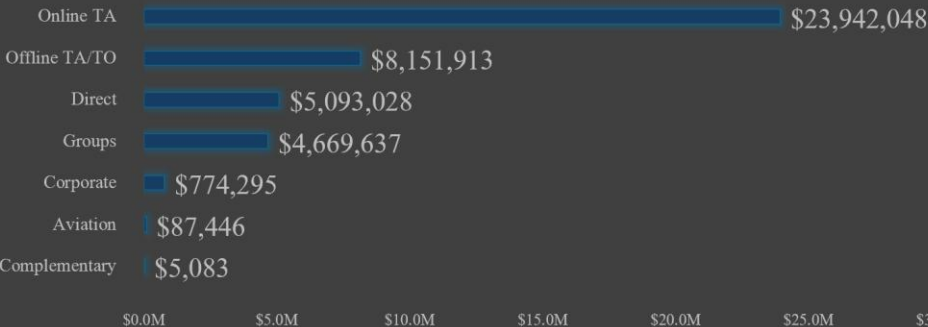
Total Booking

119,390

Cancellation Rate

37.04%

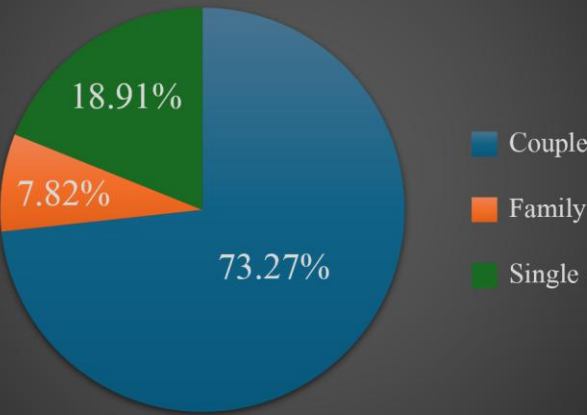
Revenue by Market Segment



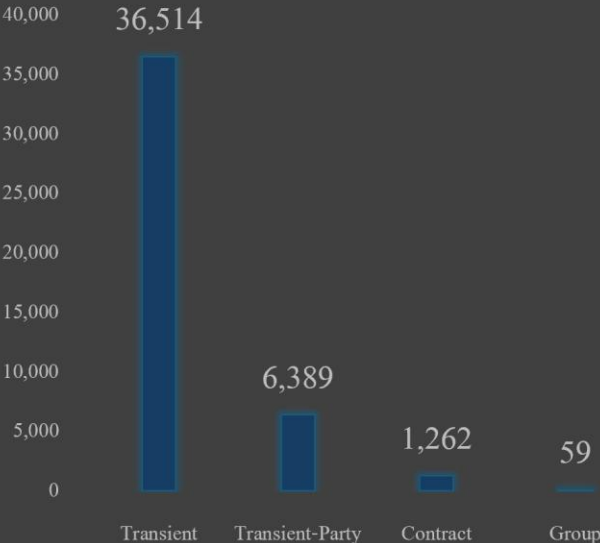
Bookings by Month



Booking by Guest Type



Cancellation by Customer Type



Monthly Bookings & Revenue

